

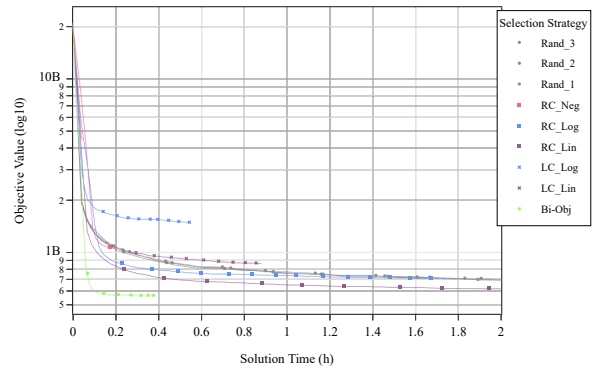
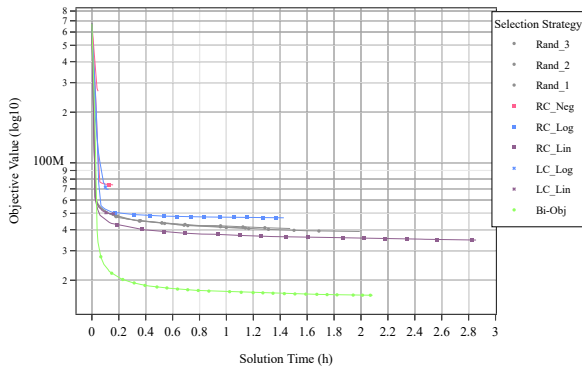
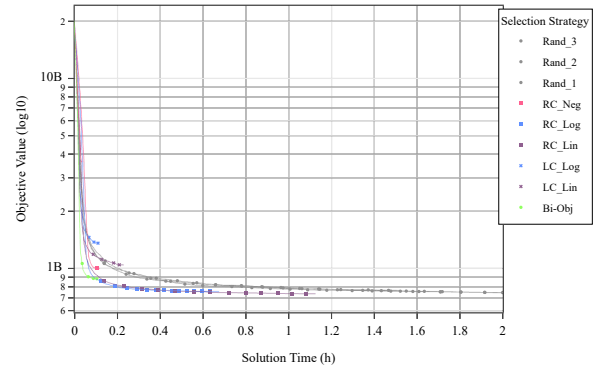
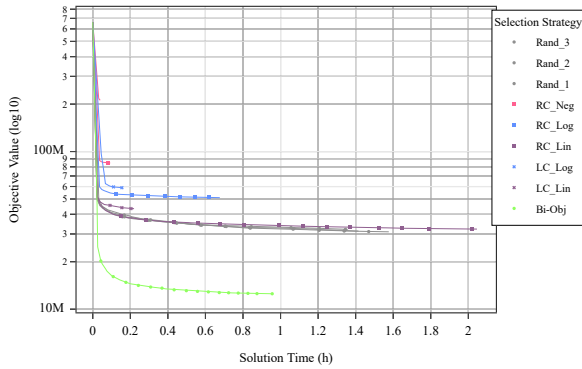
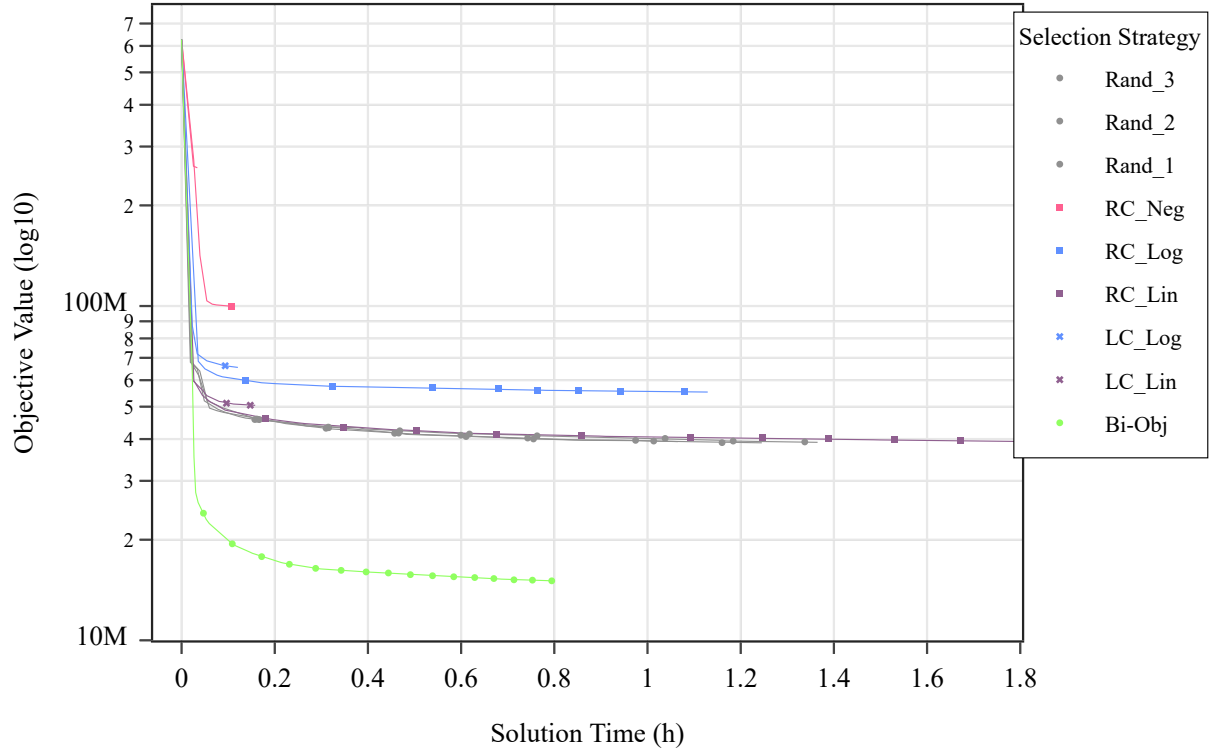
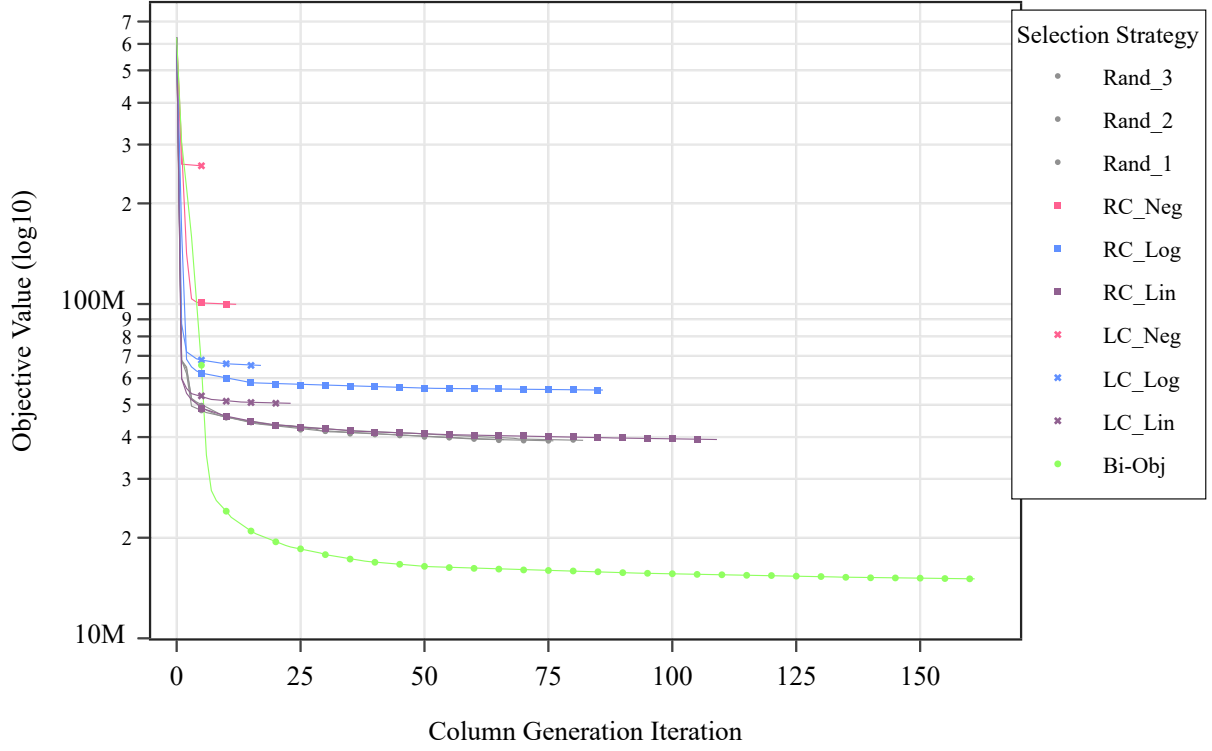
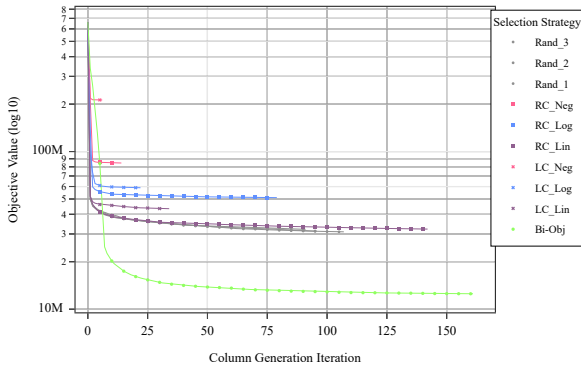


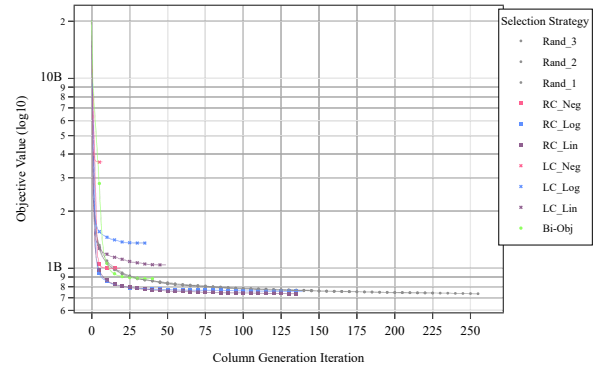
Figure 1: The LP objective value (on a $\log_{10}$ scale) plotted against the wall-clock hours for various selection strategies (configuration: Pool_Selection). Each marker represents every 10 column generation iteration.



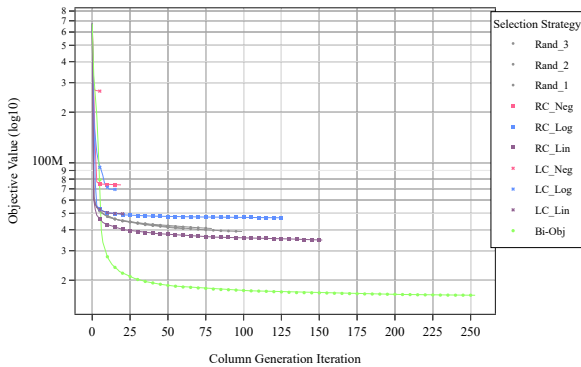
(a) Instance: A2



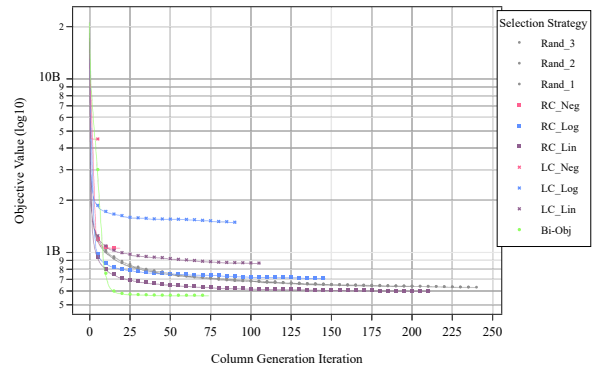
(b) Instance: A1



(c) Instance: B1

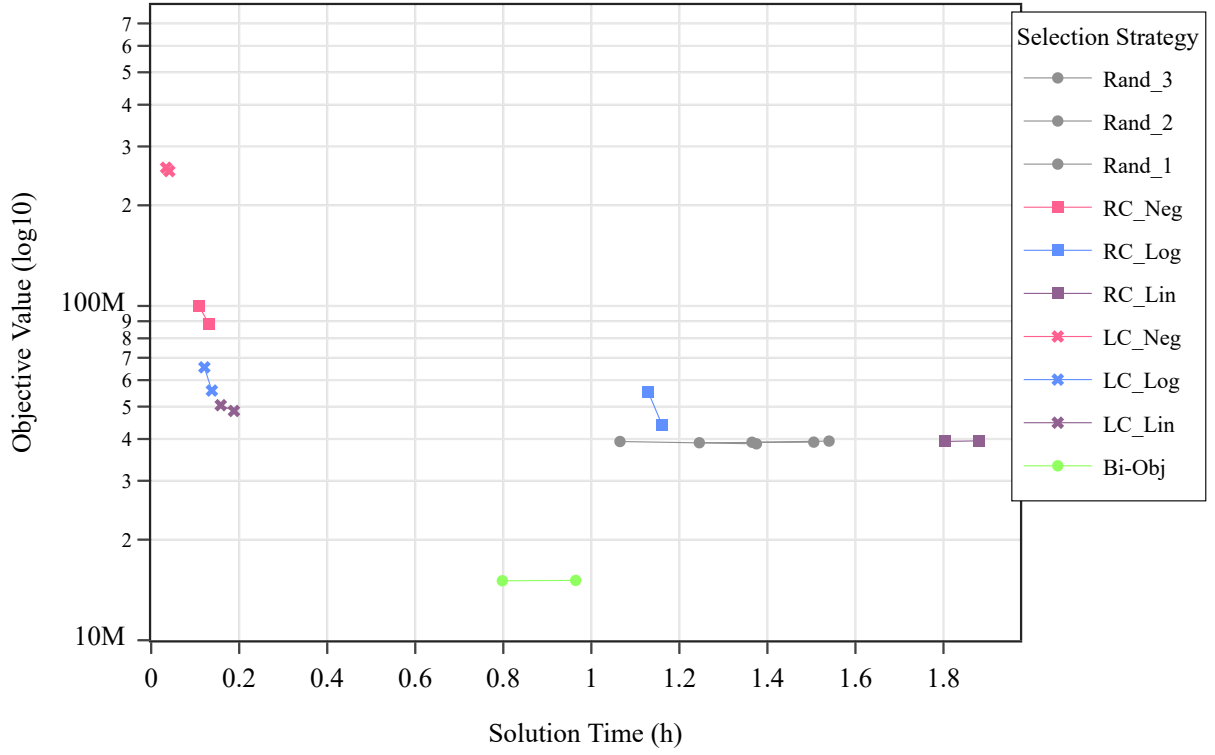


(d) Instance: A3

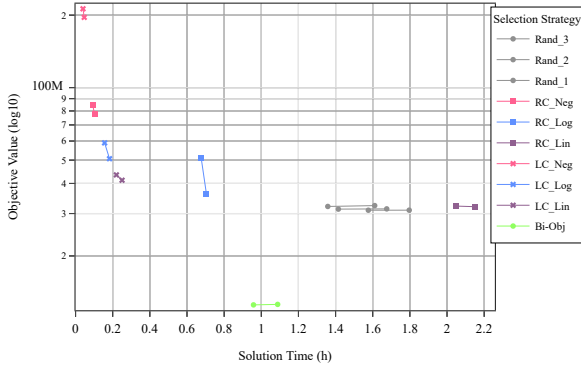


(e) Instance: B2

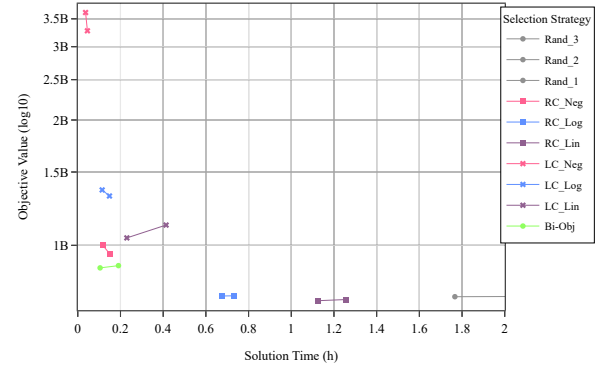
Figure 2: The LP objective value (on a $\log_{10}$ scale) plotted against the column generation iteration for various selection strategies (configuration: Pool_Selection). Each marker represents every 5 column generation iteration.



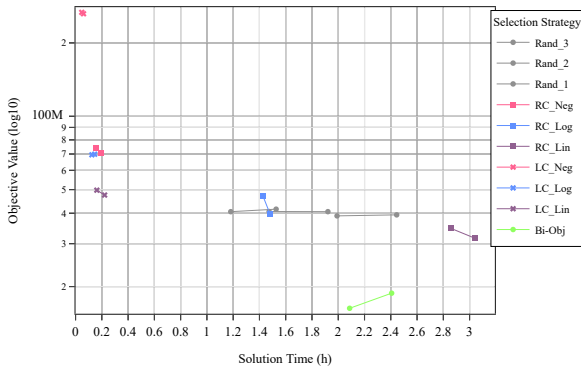
(a) Instance: A2



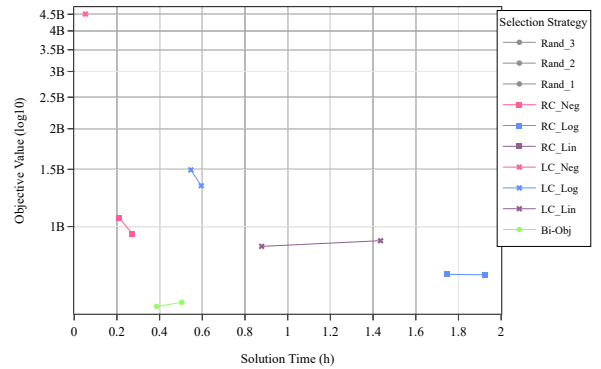
(b) Instance: A1



(c) Instance: B1

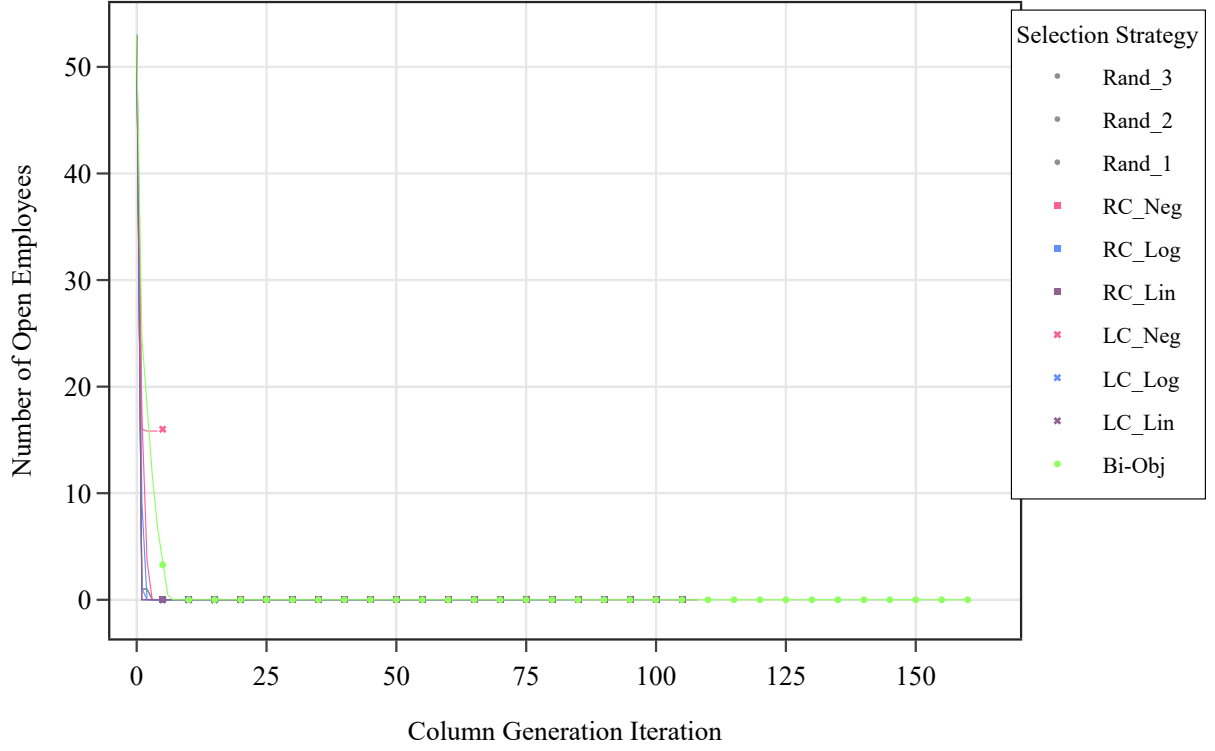


(d) Instance: A3

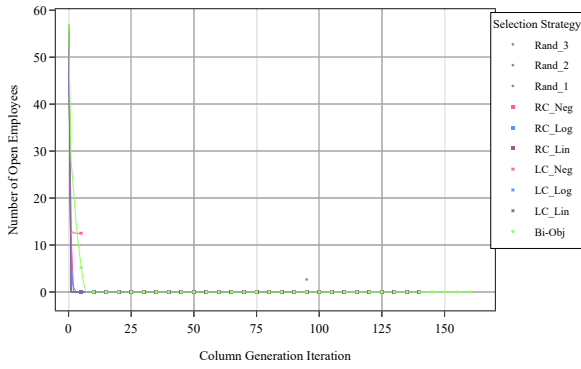


(e) Instance: B2

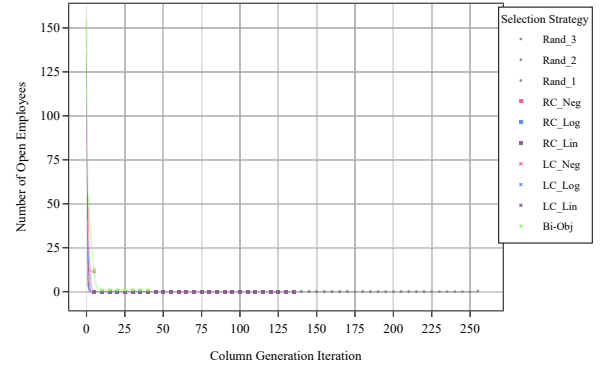
Figure 3: The final LP solution's (i.e., the best LP solution found) and the first IP solution's objective value (on a $\log_{10}$ scale) plotted against the wall-clock hours for various selection strategies (configuration: Pool_Selection). The markers are interpreted from left (LP solution) to right (IP solution) as the solution progresses in time.



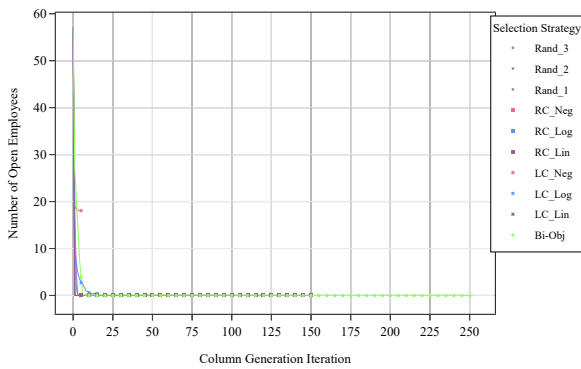
(a) Instance: A2



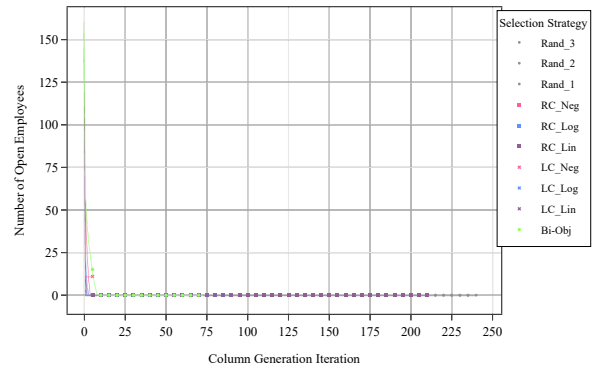
(b) Instance: A1



(c) Instance: B1

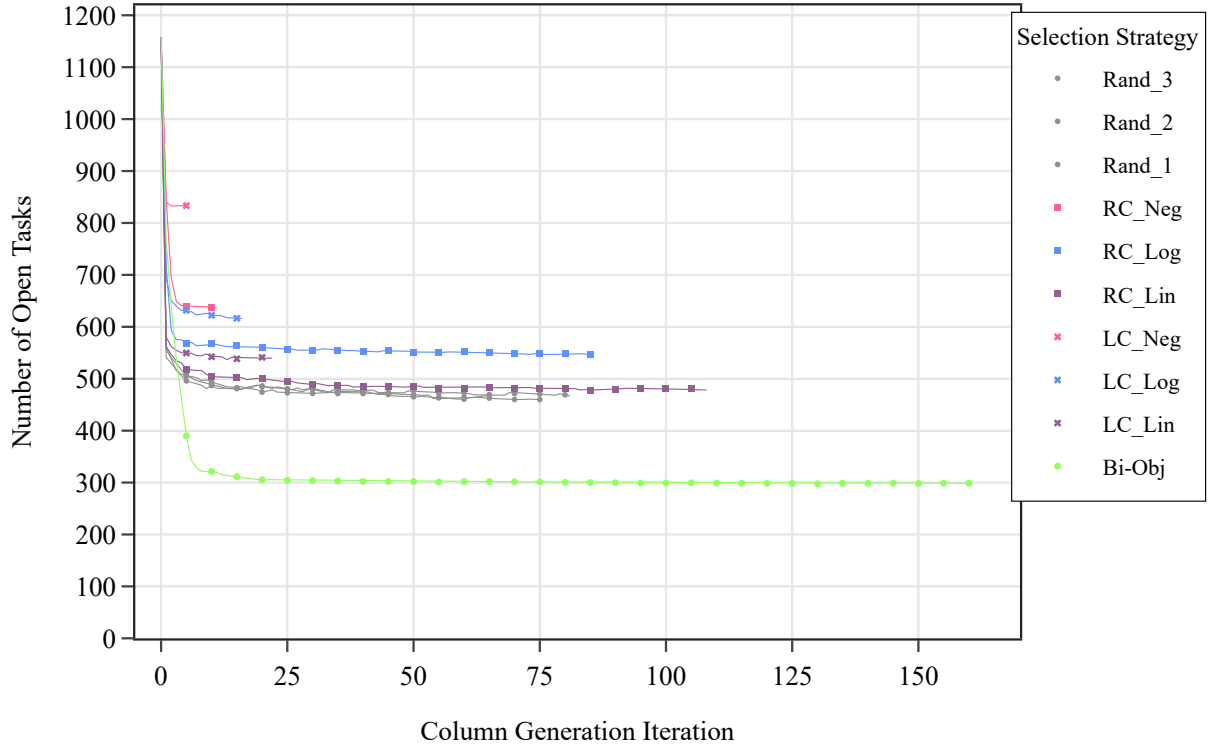


(d) Instance: A3

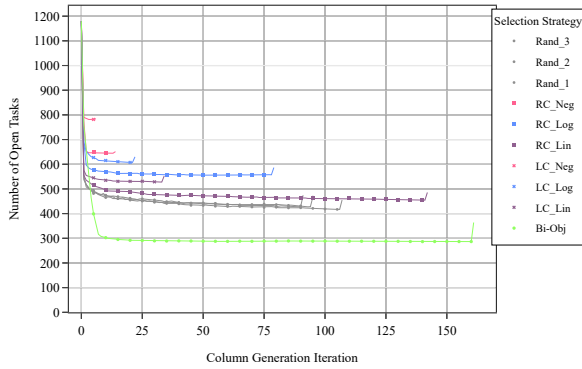


(e) Instance: B2

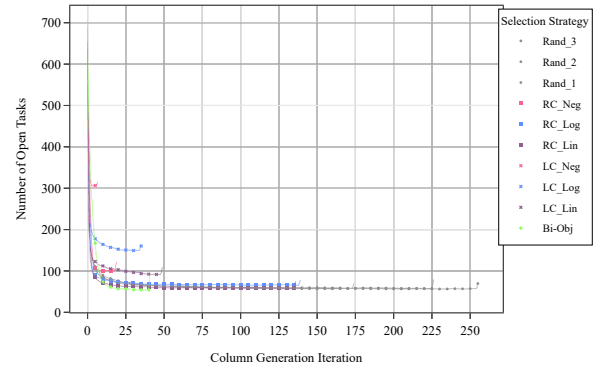
Figure 4: The number of open employees, calculated as the sum of the (fractional) artificial employee variables, plotted against the column generation iteration for various selection strategies (configuration: Pool_Selection).



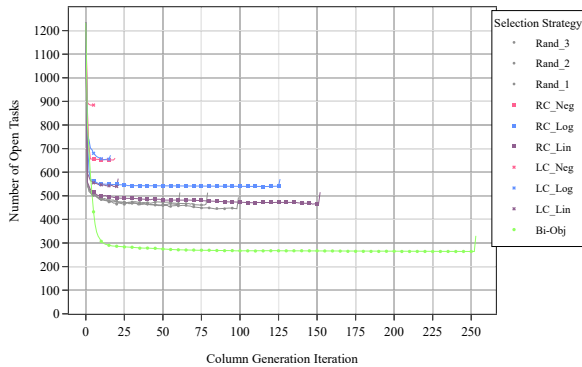
(a) Instance: A2



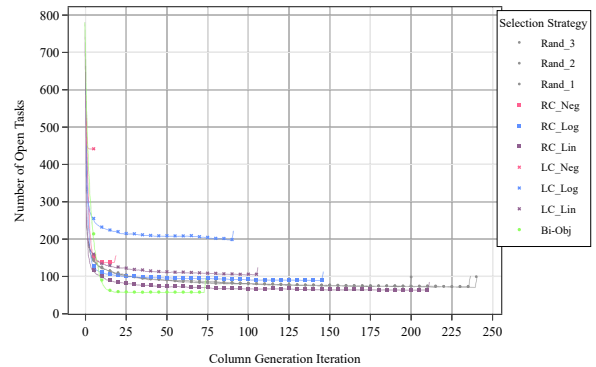
(b) Instance: A1



(c) Instance: B1

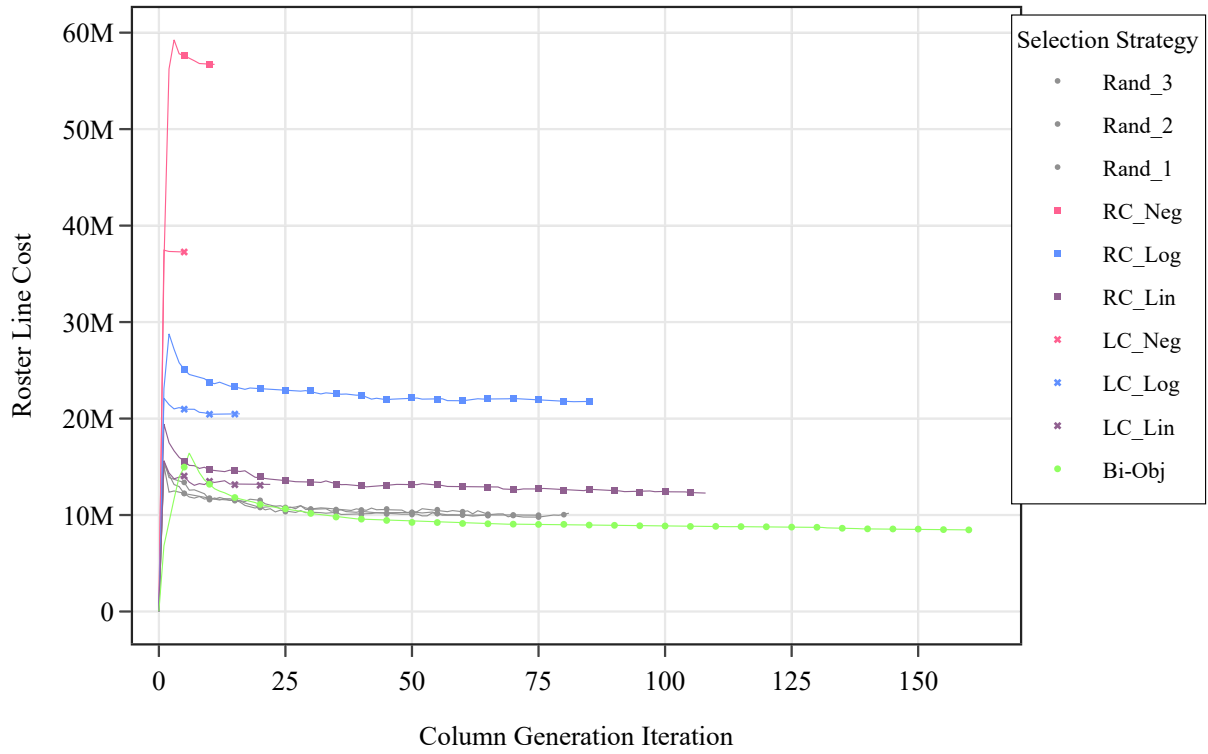


(d) Instance: A3

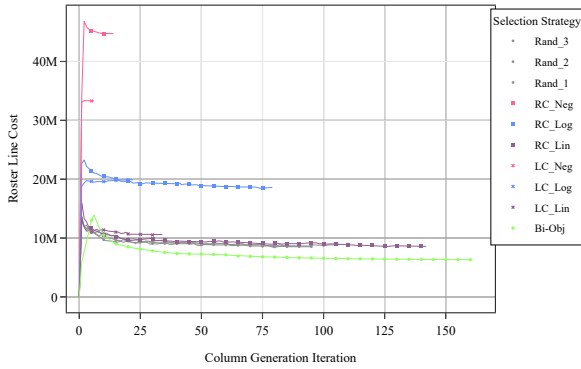


(e) Instance: B2

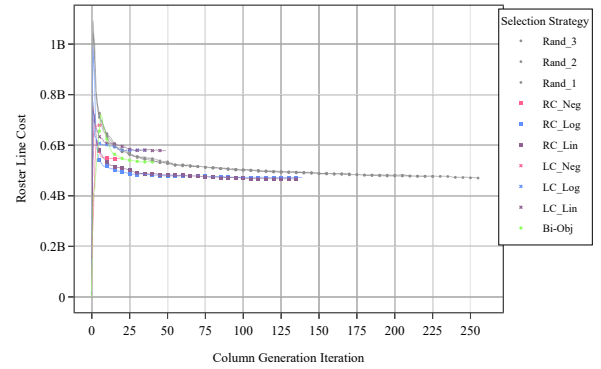
Figure 5: The number of open tasks, calculated as the sum of the (fractional) artificial task variables, plotted against the column generation iteration for various selection strategies (configuration: Pool_Selection).



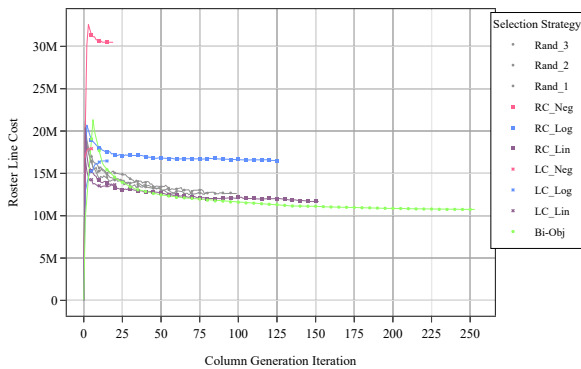
(a) Instance: A2



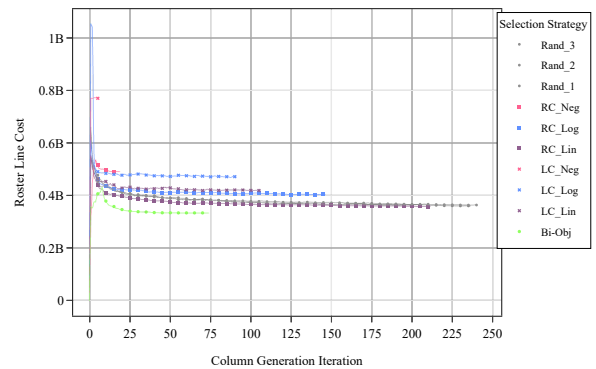
(b) Instance: A1



(c) Instance: B1

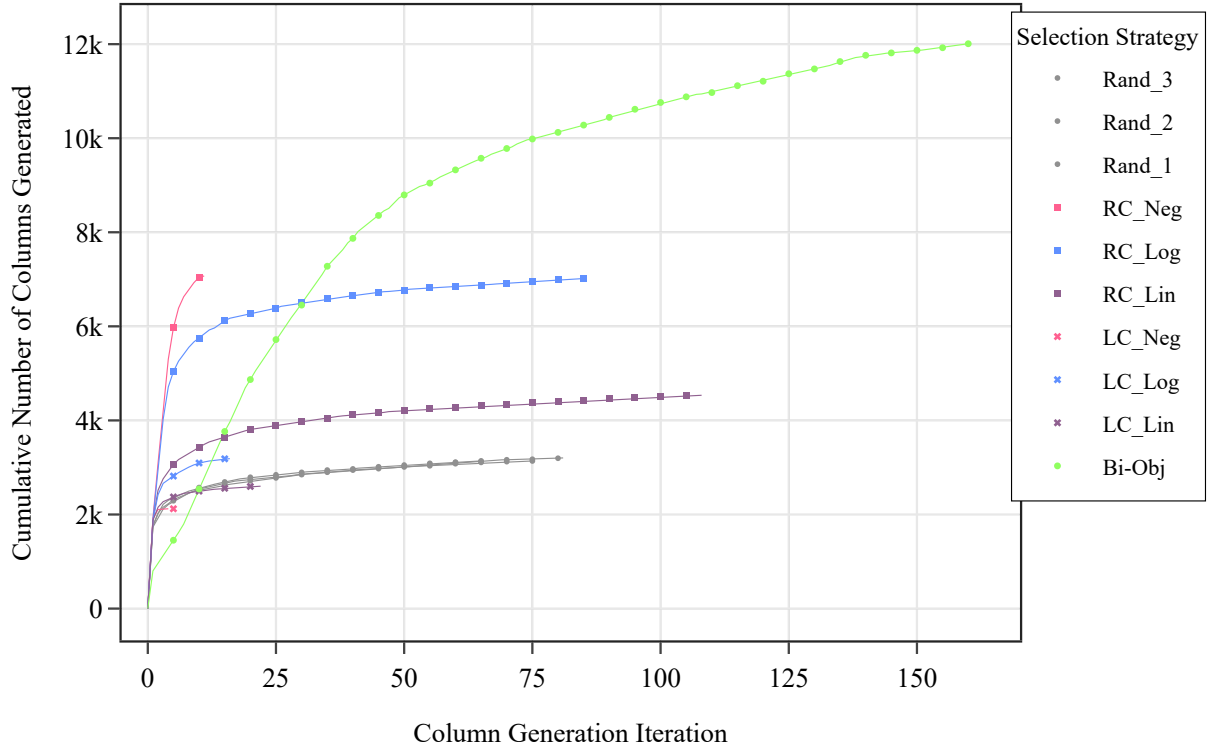


(d) Instance: A3

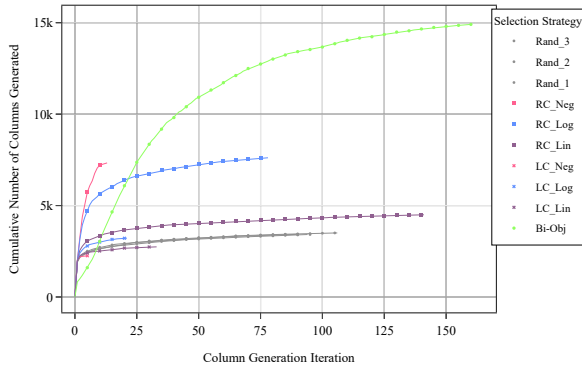


(e) Instance: B2

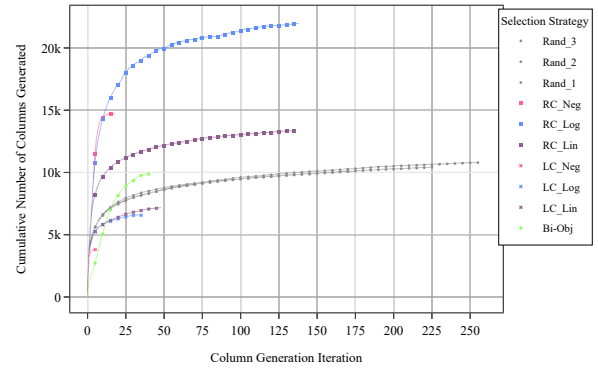
Figure 6: The total roster line cost of the solutions, calculated as the sum product of the (fractional) decision variables and individual roster line costs, plotted against the column generation iteration for various selection strategies (configuration: Pool_Selection). The line cost is composed of the soft rule violation costs.



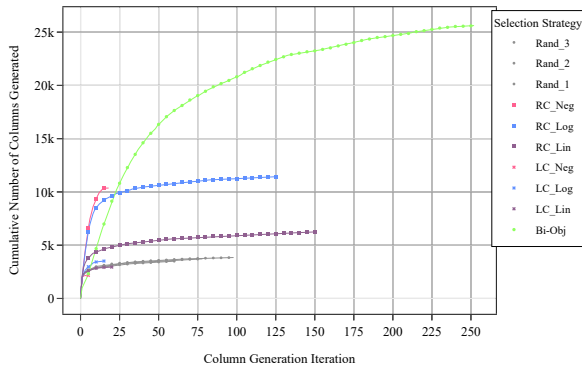
(a) Instance: A2



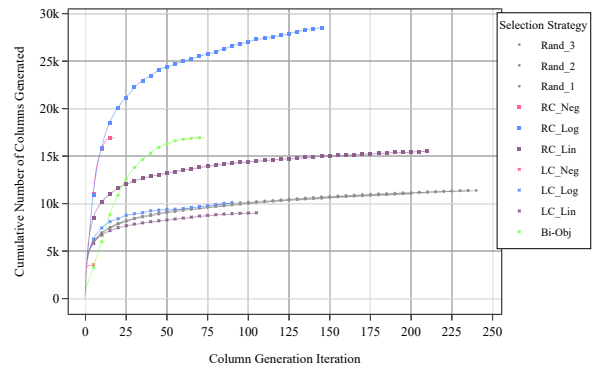
(b) Instance: A1



(c) Instance: B1

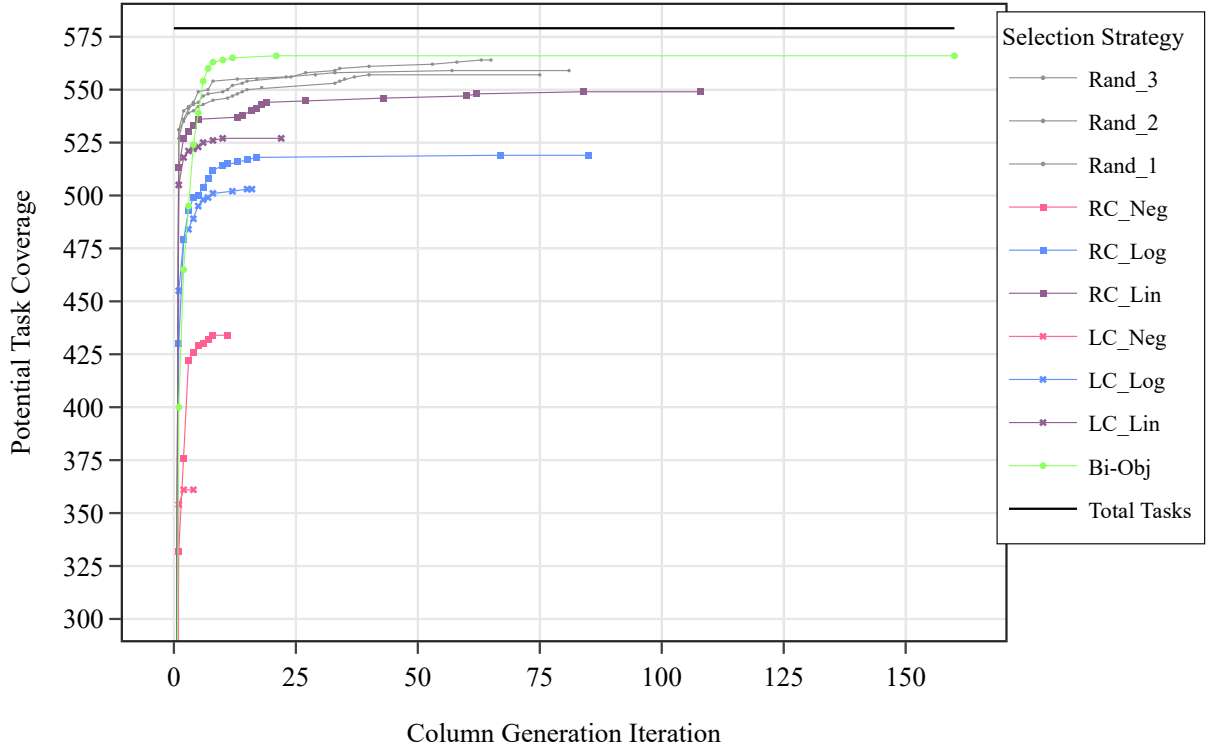


(d) Instance: A3

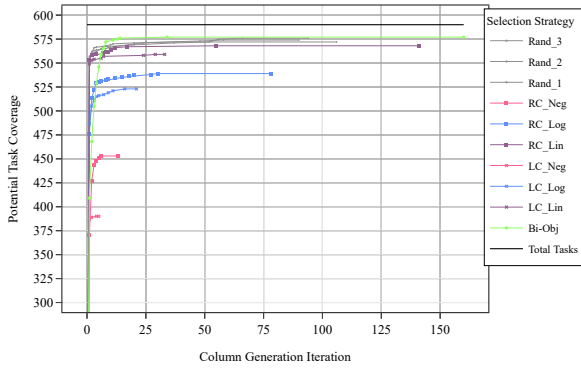


(e) Instance: B2

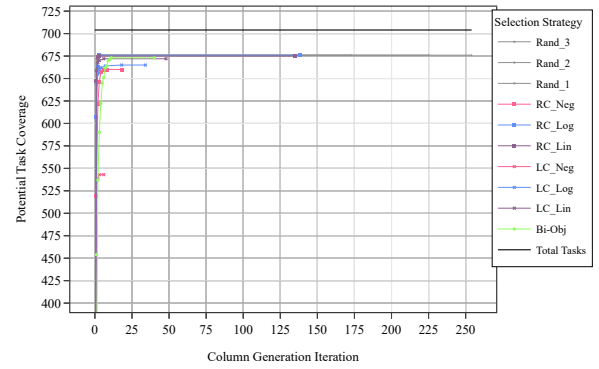
Figure 7: The cumulative number of negative reduced cost columns generated plotted against the column generation iteration for various selection strategies (configuration: Pool_Selection). The slope represents the number of negative reduced cost columns found per iteration.



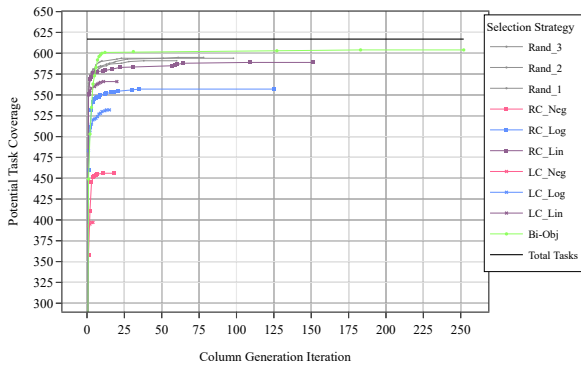
(a) Instance: A2



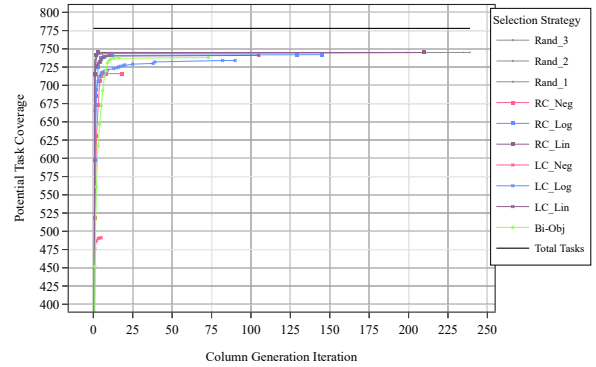
(b) Instance: A1



(c) Instance: B1



(d) Instance: A3



(e) Instance: B2

Figure 8: The potential task coverage (i.e., number of tasks that could be covered by all generated columns if the problem is formulated as a set covering problem) plotted against the column generation iteration for various selection strategies (configuration: Pool_Selection). Markers indicate iterations where changes in the cumulative number of tasks occur, with a marker repeated at the last iteration of the solution process. The black line is the total number of tasks in the problem instance.

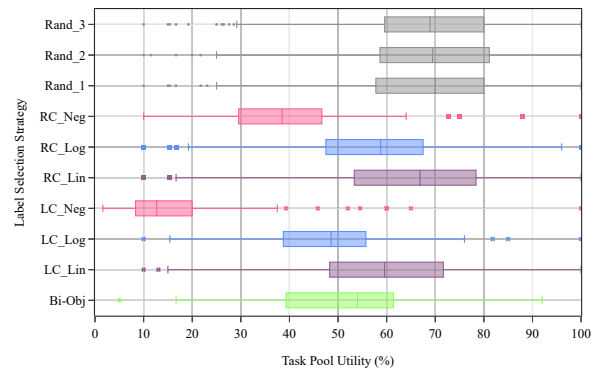
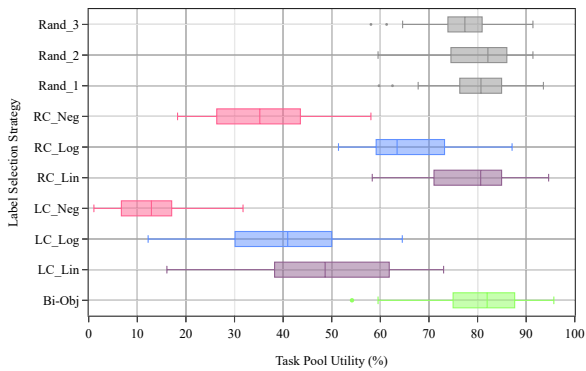
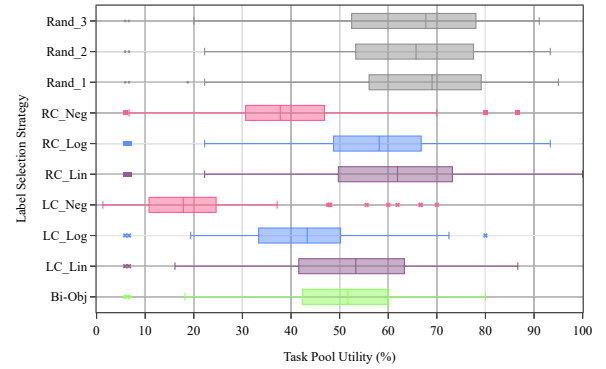
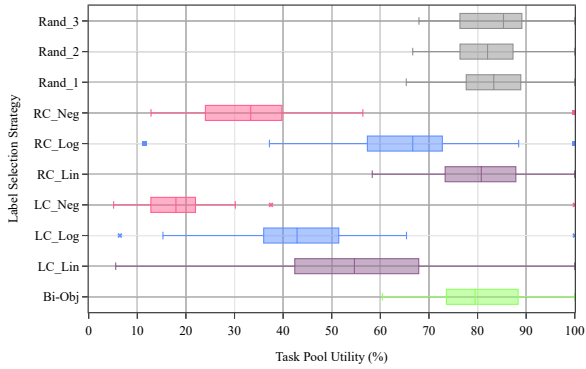
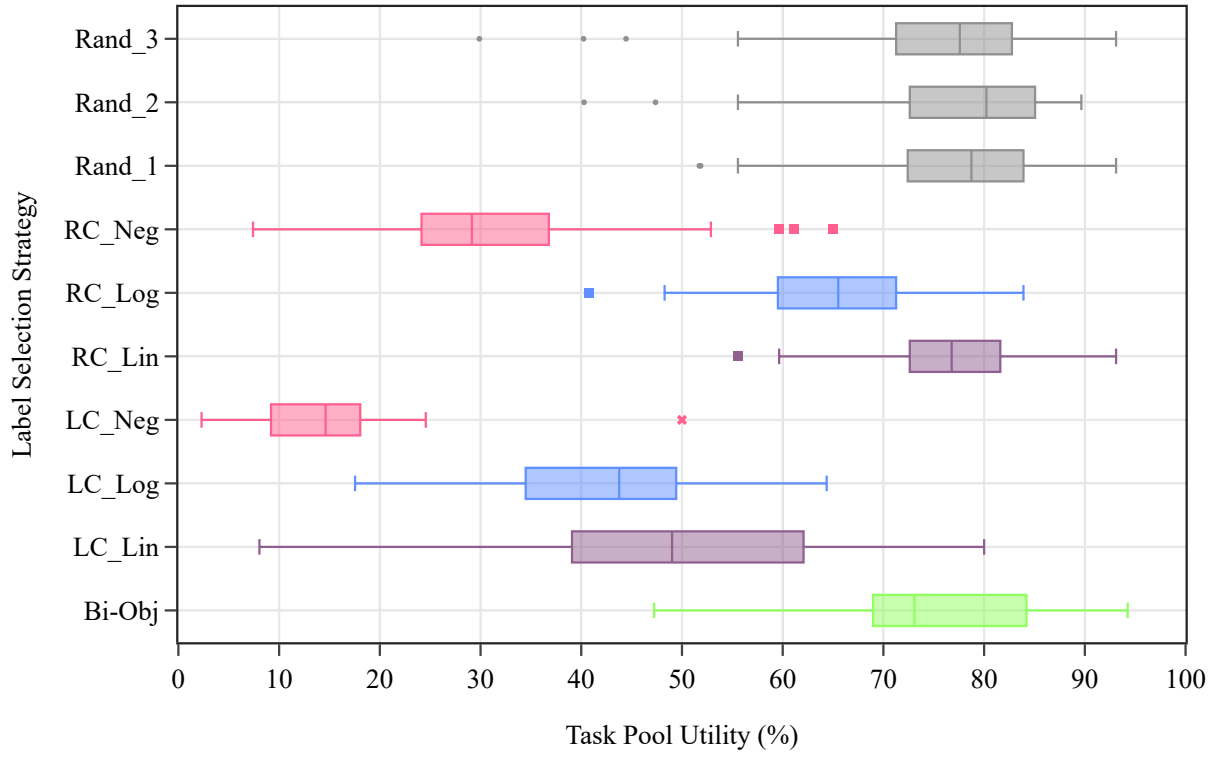
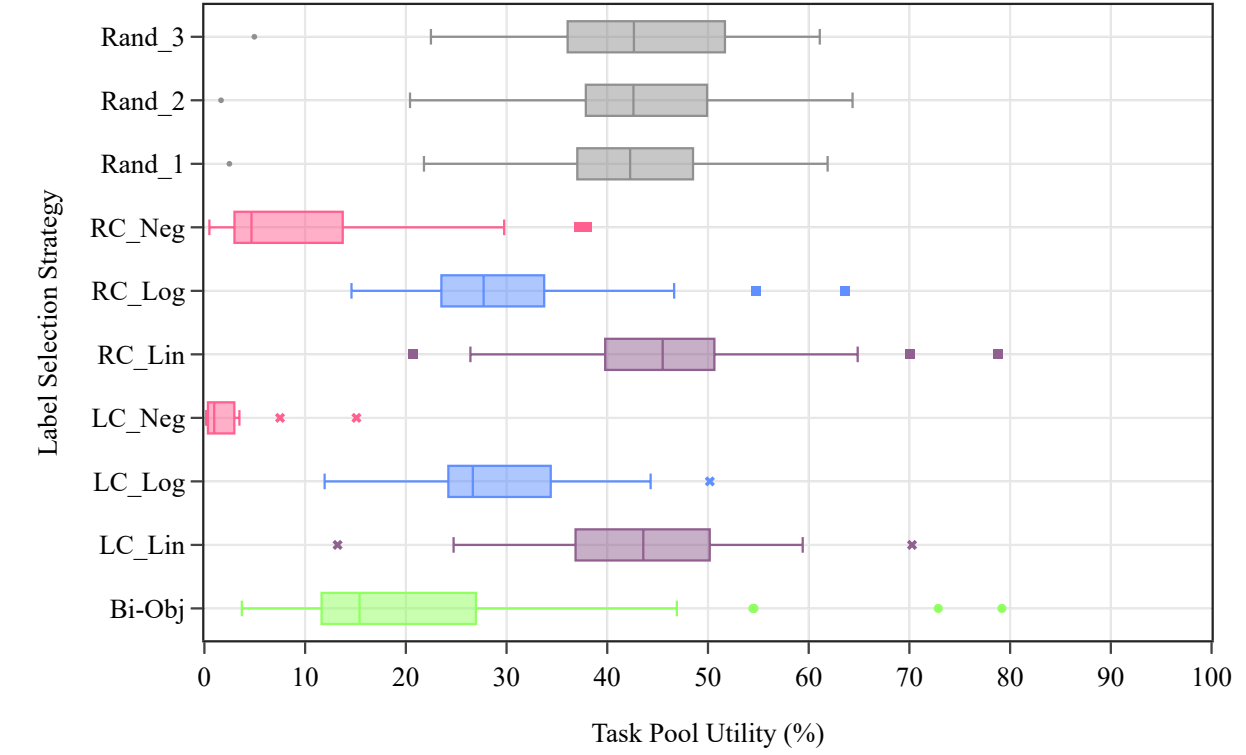
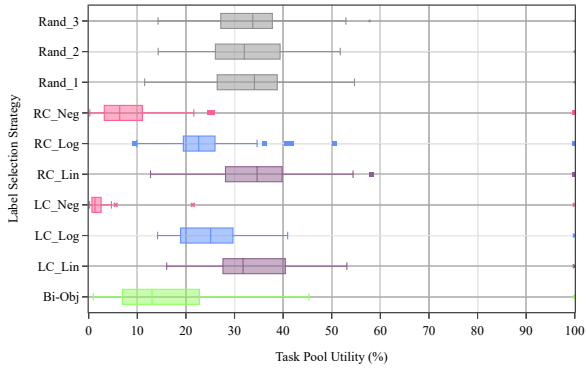


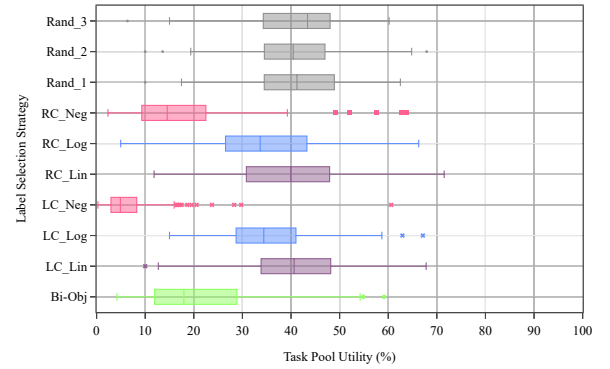
Figure 9: The spread of each employee's task pool utility for different label selection strategies (configuration: Pool_Selection). Utility is the number of unique tasks in negative reduced cost columns generated by an employee as a percentage of the total number of unique tasks in an employee's task pool. Note that for the unrestricted-task pools, each employee's task pool contains every task in the employee's eligible task set.



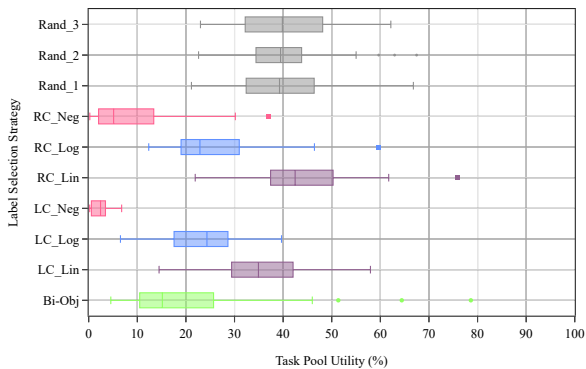
(a) Instance: A2



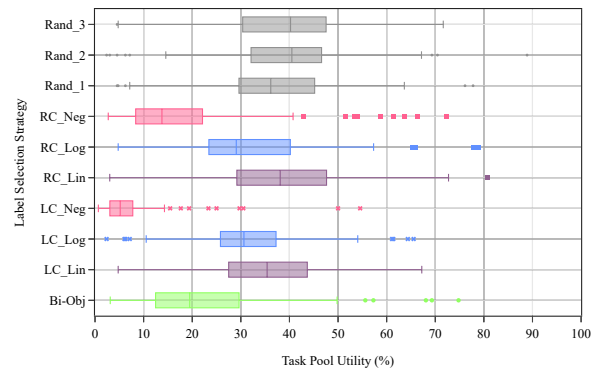
(b) Instance: A1



(c) Instance: B1



(d) Instance: A3



(e) Instance: B2

Figure 10: The spread of each employee's task pool utility for different label selection strategies (configuration: Selection). Utility is the number of unique tasks in negative reduced cost columns generated by an employee as a percentage of the total number of unique tasks in an employee's task pool. Note that for the unrestricted-task pools, each employee's task pool contains every task in the employee's eligible task set.